

PrepMatrix AI

Comprehensive Application Overview & Technology Stack Specifications

1. EXECUTIVE DESCRIPTION

PrepMatrix AI is an AI-powered study planning, practice, and progress intelligence workspace for students across school, college, and specialized academic tracks. It combines each learner's academic profile, subjects, chapter load, difficulty, exam date, and daily completion data to generate an adaptive study path, reveal weak areas, and measure exam readiness.

Within one authenticated workspace, learners can manage goals and reminders, capture notes, discover syllabus-aware materials, generate topic quizzes, take secure exams, export reports and question papers, and ask a planner-aware study companion questions through text, voice, image, or PDF inputs. Cloud-backed workspace sync keeps the study system connected across sessions, while themes and accessibility-oriented controls personalize the experience.

2. TECHNOLOGY STACK SPECIFICATIONS

Frontend Ecosystem

- React 19 - Component-driven SPA with lazy routes.
- Vite 8 - Fast development and optimized builds.
- React Router 7 - Protected study and settings routes.
- Recharts - Progress, workload, readiness, and trend charts.
- Lucide + CSS3 - Responsive icons, themes, glass cards, and motion.
- Three.js / R3F - Interactive animated study visuals.

Backend Ecosystem

- Node.js + Express 5 - REST APIs, SPA delivery, validation, and orchestration.
- MongoDB driver - Indexed user, plan, note, quiz, and exam data.
- PBKDF2 + sessions - Salted hashes, expiry, and scoped API access.
- Email verification - OTP delivery through Nodemailer or Resend.
- Push + PDF services - VAPID alerts and bounded PDF parsing.

Data, AI & Integrations

- Groq Cloud AI - Planner-aware chat, quiz, exam, and paper generation with text, image, and bounded PDF context.
- Browser capabilities - Speech recognition, wake phrases, spoken prompts, service-worker alerts, and push where supported.
- Persistence & export - MongoDB workspace sync plus jsPDF/html2canvas reports, timetables, mind maps, results, certificates, and JSON backup.

3. APPLICATION PAGES & COMPONENTS

Authentication & Learner Profile

Registration and login create a private study workspace. Learners can store their academic stage, class or degree, curriculum or track, institution details, profile image, and account preferences; OTP and password checks protect sensitive profile changes.

Dashboard (Home)

A connected overview of overall completion, first pending task, weakest subject, exam readiness, daily momentum, goals, insights, and upcoming work. Completion rewards mark daily wins, full-plan progress, and the 80% exam unlock.

Subjects & Academic Context

Build a subject library with chapter counts and Easy, Medium, or Hard difficulty. Class, board, stream, degree, and specialization context carries into planning, materials, quizzes, exams, and assistant explanations.

Smart Planner & Timetable

Generate a focused schedule from subjects, exam date, and Balanced, High priority, Revision-heavy, or Rapid coverage strategies. Mark tasks complete, recover backlog, rebalance overloaded days, undo adjustments, start a new plan, and export the timetable as PDF.

Goal & Reminder Center

Create dated goals, scheduled reminders, quick to-dos, daily study targets, and weekly review targets. Browser alerts, sounds, postponement actions, and scheduled 6 PM push nudges where supported keep incomplete work visible.

Analytics & Readiness

Explore completion trends, XP, levels, badges, topic timelines, the focus map, task distribution, study rhythm, subject performance, weekly velocity, exam-readiness signals, and focused next-step insights.

Notes & Doubt Board

Capture chapter summaries, unresolved doubts, and left-over topics by subject. Search and filter the board, track note status, and bridge unfinished topics back into planning and revision workflows.

3. APPLICATION PAGES & COMPONENTS - CONTINUED

AI Study Companion, Chat History & Voice

A planner-aware assistant adapts explanations to the learner profile and current task metrics. It preserves chat sessions, accepts images and PDFs, supports wake phrases and navigation commands, and can be opened through the animated sidebar study pet.

Interactive Quizzes

Generate 5- or 10-question topic MCQ sets with learner-stage and subject context. Submit answers, review scores, retain attempt history, export an attempt as PDF, and identify topics needing more practice.

Secure Exam Workspace

At 80% planner completion, generate a 40-question, 60-minute exam with server timing, autosave, flags, fullscreen monitoring, and server scoring. Learners get at most two starts in a rolling 24-hour window; submission locks for 15 minutes, and a fourth violation auto-submits. Results unlock after 72 hours, with certificates for qualifying scores.

Question Papers & Offline Practice

Create saved, printable AI question papers with configurable subjects, marks, difficulty, language, internal choice, question shuffling, model answers, marking schemes, and answer keys. A persistent focus/break timer supports offline solving sessions.

Study Materials

Receive subject and chapter-aware pathways to videos, articles, references, and targeted searches based on academic context and progress. Save, search, open, and remove bookmarks in a personal revision library.

Reports, Exports & Worktree Mind Maps

Generate a multi-page planner intelligence report with profile, completion, subject watchlist, quiz performance, bookmarks, study rhythm, and next actions. Export timetables, mind maps, papers, results, and certificates, or back up and restore the workspace as JSON.

Settings, Appearance & Product Guide

Configure themes, backgrounds, brightness, glass effects, accent colors, typography, card density, cursor style, sounds, notifications, wake mode, study targets, and profile details. Data controls support import, export, reset, and verified account deletion; the About guide explains the complete workflow.

3. APPLICATION PAGES & COMPONENTS - CONTINUED

Academic Profile Workspaces

Keep separate academic contexts under one account. Switching profiles scopes subjects, plans, learning records, code drafts, and other study data to the active learner profile.

Start Learning Notebooks & Mastery

Turn study material, including PDFs, into saved revision notebooks with notes, questions, and concept maps. A guided learn, recall, practice, and mastery flow tracks misconceptions and schedules review.

CodeMatrix Compiler

Write and run Python, JavaScript, SQL, HTML/CSS/JavaScript, C, C++, and Java in an eligible learner workspace. Keep profile-scoped drafts, use terminal input, inspect errors and previews, and request optional AI error hints.

Quiz Battles

Create a private 10-question quiz battle or join a friend with an invite code. Each learner gets a timed attempt with answer autosave; results and question review become available after both finish or the battle closes.

Placement Preparation

Use a saved notebook or a custom topic brief to generate role-focused interview preparation. Review topic guidance and preparation plans, save history, and carry coding practice into CodeMatrix where relevant.

Resume Builder & Analyzer

Build and export a styled resume, keep saved versions, and review a builder draft or uploaded resume against a target role or job description. Feedback highlights skill gaps and specific improvements.

Answer Coach

Upload clearly numbered handwritten answer images or a PDF for a saved question paper. Receive provisional step marks, first-error feedback, targeted practice, and rechecks; weak topics can become planner tasks.

3. APPLICATION PAGES & COMPONENTS - CONTINUED

Kids Learning Mode

Younger learners get an age-aware learning space with adventure maps, short games, daily missions, a pet tutor, and a parent corner. Parent-guided access protects more advanced routes.

Nearby Study Spots & Revision Circles

Find libraries and other local study spots, save places, and open directions. Nearby revision circles support joining or hosting group study sessions and connecting them to a study plan.

Medical Training Lab

Eligible health-science learners can generate educational conceptual reasoning modules, practice checks, and training sequences from course context, then save or revisit the generated work.

Learning Insights & Review

Mastery maps, active recall records, review queues, and subject progress views surface what is due next. Planner and analytics views connect those signals to daily momentum and follow-up work.

AI Credits & Material Consent

Usage-based AI features show account credit availability and applicable costs. Start Learning presents a privacy disclosure before sending uploaded material to AI services for notebook generation.

4. TECHNOLOGY STACK ADDITIONS

Data, AI & Browser Runtime

- Gemini handles native-PDF Start Learning requests first, with a Groq fallback; other Groq-powered study features remain available.
- CodeMirror editing and isolated browser execution use Pyodide, QuickJS, sql.js, WebAssembly Clang, and Doppio for supported languages.
- MongoDB-backed services synchronize profile-specific learning, coding, battle, resume, and assessment records across sessions.